Root Finding & Interpolation

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Numerical Methods

Universe

- Root Finding: \{methods, best_method, best_root_degC, best_residual\}
- Interpolation: \{cubic_spline_max_error, linear_max_error, lagrange_max_error, max_spline_vs
- Integration: \{trapezoidal, simpson, gaussian_quad\}

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

Given 8 measured data points of temperature (0--700 degC) vs. thermal expansion coefficient alpha(T), which increases nonlinearly from ~9 to ~16 x10^-6 /degC:

1. **Root finding**: Find the temperature where alpha(T) = 12.0 x10^-6 /degC by solving g(T) = spline(T) - target = 0
2. **Method comparison**: Compare 4 root-finding methods -- bisection, Newton (finite-difference), secant, and Brent
3. **Interpolation comparison**: Fit the data using linear, cubic spline, and Lagrange interpolation
4. **Numerical integration**: Integrate alpha(T) over [100, 500] degC using trapezoidal, Simpson, and Gaussian quadrature

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Bisection**: Guaranteed convergence (linear rate), requires bracket

(2) Newton's method**: Quadratic convergence near root, needs derivative (approximated via

(3) Secant method**: Superlinear convergence (~1.618), no derivative needed

(4) Brent's method**: Combines bisection reliability with superlinear speed *the gold standard*

(5) Cubic spline**: C2-continuous interpolation that passes exactly through data points

(6) Gaussian quadrature**: Exact for polynomials up to degree 2n-1 with n points

Approach

Suggested approach: Cubic Spline + Brent/Bisection/Newton/Secant root finding

Complexity class:

Available tools: Cubic Spline + Brent/Bisection/Newton/Secant root finding

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Cubic Spline + Brent/Bisection/Newton/Secant root finding
Time:	0.0113s

Solution Details

Solution computed successfully.

Verification

✓	All Methods Find Root	Passed
✓	Roots Agree	Passed
✓	Residual Small	Passed
✓	Root In Range	Passed
✓	Spline Passes Through Data	Passed
✓	Integration Methods Agree	Passed
✓	Integration Positive	Passed
✓	Brent Fewest Or Tied	Passed

3. Interpretation

The Question

Given 8 measured data points of temperature (0--700 degC) vs.

The Answer

A feasible solution was found.

What This Means

- Root (Brent): ~244 degC where alpha(T) = 12.0 x10^-6 /degC
- Method agreement: All 4 methods agree to within 0.1 degC
- Brent iterations: Fewest (or tied) compared to bisection
- Residual: |g(root)| < 1e-8 for Brent's method
- Integration: All 3 quadrature methods agree within 1%
- 8 verification checks: All pass

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio